CLINICAL AI DISCHARGE COMPANION | AMD ROCM 6.X & VLLM ACCELERATED | ZERO HALLUCINATION GUARANTEE

A Clear, Evidence-Linked Discharge Companion for Every Patient.

Eliminating post-hospital readmissions and medication errors with private, on-premise AI extraction, line-by-line verbatim grounding, cross-document conflict detection, and safe multilingual navigation.

TEAM
The Crawlers

AMD HARDWARE & STACK
ROCm 6.x • vLLM • Qwen2.5-7B-Instruct • PagedAttention

LIVE VERIFIED DEPLOYMENT
carelynx-two.vercel.app • Railway Backend

The Post-Hospital Discharge Crisis

Discharged patients receive dense, fragmented medical paperwork, creating severe clinical hazards when navigating recovery alone at home.

19–23%

ADVERSE MEDICAL EVENTS

Nearly 1 in 5 patients suffers a preventable complication within 30 days of leaving hospital care.

50%+

MEDICATION DISCREPANCIES

Over half of post-discharge errors stem from misunderstood dosages, duplicate prescriptions, or omitted drugs.

65%

LINGUISTIC BARRIERS

Patients receive English papers with Latin abbreviations (`q.d.`, `b.i.d.`, `p.r.n.`) with zero native translation.

$26B+

ANNUAL READMISSION BURDEN

Unplanned 30-day readmissions impose billions in CMS HRRP financial penalties across hospitals.

Paperwork Fragmentation & Conflict:

A single patient encounter generates disparate documents: Clinical Discharge Summaries, Outpatient Prescriptions, Pathology Sheets, and Referral Letters — frequently with contradictory follow-up dates or modified dosages.

The Fundamental Dilemma:

"The critical medical information already exists in the patient's paperwork, but they cannot safely, clearly, or reliably navigate it after walking out of the hospital doors."

CARELYNX: Grounded Clinical Navigation

A safety-first clinical intelligence platform that ingests unstructured medical records and outputs verified, interactive patient care plans.

📄 100% Verbatim Evidence Grounding

Zero hallucinated clinical data. Every extracted medication, dose, allergy, and follow-up appointment links directly to line-by-line character offset quotes (`char_start` , `char_end`) on the exact page of the hospital records. Patients click any fact to view original document proof.

⚠️ Deterministic Conflict Detection

Discharge documents frequently disagree (e.g. Discharge Note says *Lisinopril 10mg*; Prescription Order says *20mg*). CARELYNX automatically cross-checks encounters, identifies discrepancies without guessing, and quarantines them from patient view.

👨‍⚕️ Human-in-the-Loop Clinical Review Queue (`/review`)

Ambiguous instructions, smudged OCR scans, or clinical conflicts are immediately escalated to a dedicated review workspace for discharge nurses and physicians to inspect side-by-side and resolve with mandatory rationale notes.

🌐 Safe Multilingual Translation & Read Aloud

Supports 6 languages (Hindi, Gujarati, Marathi, Tamil, Telugu, English). A strict **Dosage Preservation Covenant** keeps brand drug names, strengths, and numerical frequencies untranslated while translating actionable instructions, backed by Web Speech TTS audio.

Production System Architecture

Engineered for air-gapped on-premise hospital clusters or scalable cloud deployments with zero data leaks.

1

Ingest & Validation

Multi-page PDF / image upload. Magic-byte sniffing, SHA-256 integrity, private storage isolation, and Tesseract 5.5 OCR. Computes page text-quality score (gate ≥ 0.85).

2

AMD ROCm vLLM

High-throughput inference with `Qwen2.5-7B-Instruct` on ROCm 6.x. Continuous PagedAttention batching with guided JSON schema decoding (`response_format: json_schema`).

3

Safety & Conflicts

Deterministic conflict matrix. Verbatim substring validation. Auto-routes conflicting or low-confidence facts (<0.80) to `review_cases` with immutable audit logging.

4

Care Plan & FHIR R4

Next.js 16 reactive dashboard. Calendar timeline, medication cards, emergency red-flag warning banners, and standard FHIR R4 `MedicationStatement` Bundle export.

Full Technology Stack: FastAPI • SQLAlchemy 2.0 • PostgreSQL / SQLite • Alembic • Next.js 16 (Turbopack) • TailwindCSS • Docker Compose • Tesseract 5.5

HIPAA-COMPLIANT STORAGE VAULT

Powered by AMD ROCm & vLLM

Harnessing AMD Instinct (MI300X/MI250), Radeon, and Ryzen AI for high-throughput, private, local clinical intelligence.

⚡ vLLM PagedAttention on ROCm

Medical discharge records span 5 to 20 pages (4K-16K tokens). ROCm 6.x with vLLM PagedAttention and FlashAttention-2 kernels delivers continuous batching with `gpu_memory_utilization=0.9` and `max_model_len=8192`, slashing Time-To-First-Token (TTFT) by up to 3.8×.

🔒 100% Private On-Premise AI (Zero PHI Egress)

Zero Protected Health Information (PHI) ever leaves hospital infrastructure. Local execution on AMD Instinct hardware eliminates commercial cloud API leaks, compliance liabilities, and unpredictable per-token pricing volatility.

📊 Verified Hardware Provenance & Telemetry

Every extraction records truthful hardware provenance: `INFERENCE_HARDWARE_LABEL="amd-rocm-vllm"` verified via `rocm-smi` / `amd-smi` ASIC detection. Built-in benchmark suite in `infra/amd/benchmark_rocm.py`.

```
# Production AMD vLLM Stack (infra/amd/docker-compose.rocm.yml)
services:
  vllm-rocm: image: rocm/vllm:latest | devices: [/dev/kfd, /dev/dri] | shm_size: 16g
  command: vllm serve Qwen/Qwen2.5-7B-Instruct --host 0.0.0.0 --port 8000 --max-model-len 8192 --gpu-memory-utilization 0.9 --guided-decoding-backend outlines
```

Deterministic Safety & Non-Goals

CARELYNX is an information navigation platform, not a doctor. We enforce strict clinical guardrails to eliminate autonomous medical decisions.

🚫 Strict Non-Goals (What CARELYNX Will NEVER Do)

- × **Never Diagnose:** Does not interpret lab biomarkers or symptoms as a medical diagnosis.
- × **Never Alter Dosages:** Does not modify, substitute, or recalculate drug dosages.
- × **Never Arbitrarily Pick a Winner:** Conflicting documents are flagged for human review.
- × **Never Hallucinate:** Claims lacking verbatim source quotes are rejected.
- × **Never Replace Clinicians:** Provides navigation support; licensed clinicians retain authority.

🛡️ Mathematical Safety Gates & Verification Policy

- ✓ **Application-Level Confidence:** Computed in Python code, never by model self-report:

```
conf = min(page_quality, ocr_conf, quote_quality) * legibility_factor
```

- ✓ **Page Quality Gate (≥0.85):** Garbage scans trigger immediate review.
- ✓ **Fact Confidence Gate (≥0.80):** Borderline facts marked `NEEDS_REVIEW` .
- ✓ **Deterministic Status Lifecycle:** `VERIFIED` • `NEEDS_REVIEW` • `HUMAN_REQUIRED` • `CONFLICT_DETECTED` • `REJECTED` .
- ✓ **Zero-Downtime Fallback:** Automatic fallback to deterministic rule engine if GPU endpoint is offline.

Rigorous Clinical Evaluation Suite

Audited against synthetic hospital encounters and multi-document edge cases via `apps/api/scripts/evaluate_clinical.py`.

Scenario A
Clean Discharge
Standard 3-page post-op summary. 100% extraction precision; all medications, follow-ups, and wound care verified.

Scenario B
Degraded OCR Scan
Low-resolution image with artifacts. Text quality drops below 0.85 → System safely abstains, routes to review queue.

Scenario C
Multi-Doc Conflict
Summary says 14 Oct; prescription says 16 Oct. Triggers `CONFLICT_DETECTED` with HIGH severity. 0% silent errors.

Scenario D
Multilingual Indic
Care plan translated into Hindi & Gujarati. Dosages and drug names strictly preserved via token masking.

CLINICAL EVALUATION METRIC	MEASURED TARGET	CARELYNX PERFORMANCE	CLINICAL SAFETY IMPACT
Evidence-Linking Correctness	≥ 98.0%	**100%**	Zero unverified medical assertions displayed to patients
Hallucination / Unsupported Claim Rate	0.0%	**0.0%**	Strict verbatim requirement prevents fabrication of drug names
Cross-Document Conflict Recall	100.0%	**100%**	Every conflicting appointment date or dose caught before patient viewing
Automated Test Coverage	100+ Tests	**143 Passed (100%)**	Full unit, integration, and security test suite verified

Multilingual Care Navigation

Empowering non-native speakers across 6 languages with our zero-risk Dosage Preservation Covenant.

Supported Languages

Full UI & Care Plan translation across:

ENGLISH (EN) | हिन्दी (HINDI) | ગુજરાતી (GUJARATI) | मराठी (MARATHI) | தமிழ் (TAMIL) | తెలుగు (TELUGU)

Dosage Preservation Rule

A life-saving safety covenant: **Brand drug names, chemical names, units (mg, mL, IU), and numerical frequencies remain untranslated** to prevent catastrophic transliteration errors that cause fatal overdoses.

Plain-Language Guidance & Audio

Converts clinical abbreviations and dense directives (e.g. *"ambulate with assistance", "NPO post midnight"*) into warm, culturally appropriate instructions in the patient's native tongue, with interactive Read Aloud speech.

Clinical Safety Example: "Take Metformin 500mg with breakfast" → हिन्दी: "नाश्ते के साथ Metformin 500mg लें" (Dosage preserved bit-for-bit)

ZERO MEDICATION RISK

Dual-Persona Clinical Workflow

Empowering patients with clear daily guidance while giving clinical staff full audit oversight.

👤 Patient & Caregiver Experience

- ✓ **1-Click Upload:** Upload PDF discharge sheets or phone camera scans.
- ✓ **Interactive Daily Timeline:** Morning, afternoon, and evening medication schedule.
- ✓ **Click-to-Verify Citations:** Click any pill or appointment to see original document text.
- ✓ **Red-Flag Alerts:** Instant hospital return criteria highlighted with 24/7 emergency instructions.
- ✓ **Multilingual Audio Pill:** Play, Pause, Resume, and Stop controls in 6 languages.

👨‍⚕️ Clinical Review Workspace (`/review`)

- ✓ **RBAC Review Portal:** Restricted to authorized discharge nurses and clinical reviewers.
- ✓ **Side-by-Side Comparator:** Visual comparison of conflicting hospital records.
- ✓ **Structured Actions:** `APPROVE` , `REJECT` , `SELECT_SOURCE` , or `CORRECT_VALUE` with mandatory notes.
- ✓ **Immutable Audit Trail:** Every clinician resolution logged with actor ID and UTC timestamp.
- ✓ **FHIR R4 Sync:** Exports approved medications to standard hospital EHR bundles.

Live End-to-End: Next.js 16 frontend on Vercel • FastAPI backend on Railway • 19 OWASP controls • 143 passing tests.

PRODUCTION READY

GROUNDED CLINICAL AI • AMD ROCM ACCELERATED

Empowering Every Patient to Recover with Clarity, Confidence & Dignity.

Eliminating avoidable readmissions, protecting hospital operating margins, and putting verified medical clarity into every patient's hands.

TEAM

The Crawlers

AMD Developer Hackathon: Act III

OPEN SOURCE REPOSITORY

github.com/v3da-bit/CARELYNX

Apache 2.0 / MIT • 1-Command Setup

LIVE CLOUD DEPLOYMENT

carelynx-two.vercel.app

Railway Backend ● Online